DermAI

Skin Disease Detection System

User Manual — Non-Technical Guide

⚠ IMPORTANT: DermAI is an AI-assisted screening tool only. It is NOT a substitute for professional medical diagnosis. Always consult a qualified dermatologist for any skin concerns.

1. What is DermAI?

DermAI is a web application that uses artificial intelligence to analyse dermoscopy images of skin lesions. You upload a photo of a skin lesion, and the system tells you what type of lesion it might be, how confident it is, and whether you should seek urgent medical attention.

DermAI classifies skin lesions into 7 categories:

Condition	What it is	Risk	Action
Melanoma (mel)	Most dangerous skin cancer	🔴 HIGH	See dermatologist immediately
Basal Cell Carcinoma (bcc)	Common but treatable skin cancer	🟠 MEDIUM	Consult dermatologist promptly
Actinic Keratosis (akiec)	Pre-cancerous dry patch	🟠 MEDIUM	Treatment recommended
Melanocytic Nevi (nv)	Normal mole	🟢 LOW	Monitor for changes
Benign Keratosis (bkl)	Non-cancerous growth	🟢 LOW	Cosmetic only
Dermatofibroma (df)	Harmless skin bump	🟢 LOW	No treatment needed
Vascular Lesion (vasc)	Blood vessel lesion	🟢 LOW	Monitor if growing

2. Getting Started — Creating Your Account

Before you can use DermAI, you need to create a free account. This takes less than 2 minutes.

1 **Open DermAI in your browser**
Type this address into any browser: http://localhost:7500 The login page will appear automatically.

2 **Click 'Sign Up'**
Click the 'Sign Up' tab on the login page.

3 **Fill in your details**
Enter a username (at least 3 characters), your email address, a password (at least 8 characters), and select your gender. Then click 'Create Account & Send OTP'.

4 **Check your email for the OTP**
DermAI will send a 6-digit verification code to your email. Check your inbox (and spam folder). The code expires in 10 minutes.

5 **Enter the OTP**
Type the 6-digit code into the box on screen and click 'Verify & Continue'.

6 **Log in**
You will be taken back to the login screen. Enter your email and password and click 'Login'.

💡 If you did not receive the OTP email, check your spam folder, then click 'Resend' to get a new code.

3. Analysing a Skin Image

Once logged in, you are on the main analysis page. Follow these steps:

1 **Prepare your image**
Take a clear centered, well-lit photo of the skin lesion. Dermoscopy images work best, but regular photos are accepted. Supported formats: JPEG, PNG. Minimum size: 200×200 pixels.

2 **Upload the image**
Click the upload area (or drag and drop your image onto it). A preview of your image will appear.

3 **Click 'Analyse Image'**
Click the green 'Analyse Image' button. The system will process your image. This may take 1–5 seconds.

4 **Read the results**
The results panel will show: • Condition name and risk level • Confidence score (how sure the AI is) • List of associated symptoms • Medical advisory and action to take • Scores for all 7 possible conditions

5 **View the heatmap**
Below the results, you will see a Grad-CAM heatmap — a coloured overlay on your image showing which areas the AI focused on when making its decision. Red/yellow areas are most influential.

6 **Give feedback (optional)**
Use the 👍 or 👎 buttons to tell us if the result seemed correct. Your feedback helps improve the system.

⚠ The confidence score shows how sure the AI is about its own prediction — it does NOT indicate medical severity. A 99% confident prediction of a LOW-risk condition is still not a medical diagnosis.

4. Understanding Your Results

4.1 Risk Levels

Every result includes a risk level badge:

HIGH — Seek medical attention urgently

MEDIUM — Schedule a dermatologist appointment soon

LOW — Monitor — see a doctor if it changes

4.2 Confidence Score

The confidence score (e.g. 0.91 = 91%) shows how certain the AI model is about its top prediction. A higher confidence means the image strongly resembles that condition in the training data.

> 💡 Even a high confidence score does not mean the diagnosis is definitely correct. Always use the result as a starting point for a conversation with a doctor.

4.3 All Scores Panel

The 'All Scores' section shows the probability the AI assigned to each of the 7 conditions. The scores sum to 1.0. If the top score is close to other scores (e.g. 0.40 vs 0.35), the image may be ambiguous.

4.4 The Heatmap

The Grad-CAM heatmap highlights which parts of your image the AI found most important for its decision:

- Red/yellow areas: High importance — the AI focused here most
- Blue/green areas: Low importance — less influential

Use the heatmap to check whether the AI focused on the actual lesion, or on irrelevant background. If the heatmap highlights the wrong area, treat the result with more caution.

5. Viewing Your Prediction History

All your past predictions are saved to your account. To view them:

1. Click your username (top right of the page)
2. Select 'My Predictions'
3. Your past analyses will appear in order, newest first

> 💡 Your images are saved securely and used only for future model improvement. They are never shared publicly.

6. Logging Out

To log out, click the 'Logout' button in the top-right corner of the main page. Your session will end and you will be taken back to the login page.

7. Frequently Asked Questions

Q: Can DermAI diagnose cancer?

No. DermAI is an AI-assisted screening tool. It can flag lesions that look similar to cancerous patterns in its training data, but it cannot diagnose cancer. Only a licensed dermatologist with a physical examination and, if needed, a biopsy can diagnose skin cancer.

Q: What type of images work best?

Clear, in-focus photos of a single lesion work best. Dermoscopy images (taken with a special magnifying device) give the most accurate results. Avoid images with shadows, blur, or multiple lesions in one photo.

Q: I uploaded an image but got an error.

Make sure your image is a JPEG or PNG file and is at least 200×200 pixels. If the image is corrupted or too small, the system will reject it. Try a different image or re-export your photo.

Q: I did not receive the OTP email.

- Check your spam/junk folder
- Wait 1–2 minutes — email delivery can be slow
- Click 'Resend' on the verification screen for a new code
- The OTP expires in 10 minutes — request a new one if it has expired

Q: Is my data private?

Your uploaded images and prediction results are stored in the application's database. Images are saved for the purpose of future model improvement only. No data is shared with third parties. The system uses JWT authentication to ensure only you can see your own prediction history.

Q: The analysis is taking a long time.

On a CPU-only machine (no GPU), analysis typically takes 1–5 seconds. If it takes longer than 30 seconds, the server may be processing multiple requests simultaneously. Wait a moment and try again.

8. Quick Reference Card

Action	What to do
Open DermAI	Go to http://localhost:7500 in any browser
Create account	Sign Up → fill form → check email → enter OTP → Login

Action	What to do
Analyse image	Upload zone → drag image or click → 'Analyse Image'
View heatmap	Scroll down on results page — coloured overlay on your image
Give feedback	👍 or 👎 buttons below the result
Log out	'Logout' button, top-right corner
Get new OTP	Click 'Resend' on the verification screen

⚠ This application is for educational and research purposes as part of DA5402 MLOps coursework at IIT Madras. It is NOT a certified medical device and must not be used as the sole basis for any medical decision.